WORKED SOLUTION

STEP I 2013, Question 10

Mechanics • Collision and Friction

STEP I 2013, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s1-q10

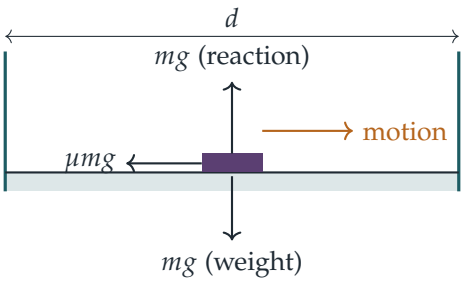
The key idea

Between collisions the only horizontal force is friction, so the puck has constant deceleration μg and the constant-acceleration formula $v^2 = u^2 + 2as$ links the speed leaving one barrier to the speed arriving at the next; Newton's law of restitution then multiplies that arrival speed by r. The result is a linear recurrence for v_i^2, and a linear recurrence $w_{i+1} = aw_i + b$ is solved by shifting to its fixed point, after which it is just a geometric sequence.

Solution

The recurrence

Model the puck as a particle moving in a straight line perpendicular to the barriers. While it is on the ice between collisions, the normal reaction is mg (no vertical acceleration), and since the puck is sliding, friction takes its limiting value μmg, opposing the motion.



By Newton's second law the deceleration is μg, constant, so for a traverse of length d starting with speed v_i (just after the ith collision) the speed u on arriving at the other barrier satisfies

$$u^2 = v_i^2 - 2\mu g d.$$

The barrier is fixed, so Newton's law of restitution says the puck leaves with speed r times its approach speed: $v_{i+1} = ru$. Squaring,

$$v_{i+1}^2 = r^2u^2 = r^2v_i^2 - 2r^2\mu gd, \qquad \text{so} \qquad v_{i+1}^2 - r^2v_i^2 = -2r^2\mu gd,$$

as required. The same argument applies to the first traverse if we write $v_0 = v$ for the starting speed, so the recurrence holds for all $i \geq 0$.

The number of traverses

Divide through by $2\mu gd$ and put $w_i = \dfrac{v_i^2}{2\mu gd}$, so that $w_0 = k$ and

$$w_{i+1} = r^2(w_i - 1).$$

"Comes to rest against a barrier after n traverses" means that the puck arrives at a barrier with zero speed at the end of the nth traverse. Then it leaves with zero speed, so $v_n = 0$, that is, $w_n = 0$ (and $w_i > 0$ for $i < n$).

Ask yourself How do we solve $w_{i+1} = r^2(w_i - 1)$?

Look for a constant solution $w_i = c$: we need $c = r^2(c-1)$, so $c = -\dfrac{r^2}{1-r^2}$, which makes sense because $r \neq 1$. Subtracting $c = r^2(c-1)$ from the recurrence gives

$$w_{i+1} - c = r^2(w_i - c),$$

so $w_i - c$ is geometric with ratio r^2:

$$w_i = c + r^{2i}(k - c).$$

Setting $w_n = 0$ gives $r^{2n}(k-c) = -c$, that is,

$$r^{2n}\left(k + \frac{r^2}{1-r^2}\right) = \frac{r^2}{1-r^2} \quad \Longrightarrow \quad r^{2n-2}(k(1-r^2) + r^2) = 1.$$

Multiplying through by r^{-2n} tidies this to

$$r^{-2n} = 1 + k(r^{-2} - 1), \qquad \text{so} \qquad \boxed{n = \frac{\ln\left(1 + k(r^{-2} - 1)\right)}{2\ln(1/r)}}$$

(for $0 < r < 1$, $\ln(1/r) > 0$, and the argument of the logarithm exceeds 1 because $k > 0$ and $r^{-2} > 1$, so $n > 0$ as it should be).

The case $r = \mathrm{e}^{-1}$

Now $r^{-2} = \mathrm{e}^2$ and $\ln(1/r) = 1$, so the formula becomes

$$n = \tfrac{1}{2}\ln\left(1 + k(\mathrm{e}^2 - 1)\right),$$

which is the required result.

The case $r = 1$

The recurrence becomes $w_{i+1} = w_i - 1$: each traverse uses up exactly $2\mu g d$ of v^2, because there is no loss at the barriers. So $w_i = k - i$, and $w_n = 0$ gives

$$n = k = \frac{v^2}{2\mu g d}.$$

This is just energy: the initial kinetic energy $\frac{1}{2}mv^2$ is all dissipated by friction over a total distance nd, so $\frac{1}{2}mv^2 = \mu mg\, nd$.

Checks

- Run the recurrence backwards with $r = \mathrm{e}^{-1}$ from $w_3 = 0$: $w_2 = 1$, $w_1 = \mathrm{e}^2 + 1$, $w_0 = \mathrm{e}^4 + \mathrm{e}^2 + 1 \approx 62.99$. The formula gives $\frac{1}{2}\ln\left(1 + (\mathrm{e}^4 + \mathrm{e}^2 + 1)(\mathrm{e}^2 - 1)\right) = \frac{1}{2}\ln \mathrm{e}^6 = 3$, as it should.
- As $r \to 1$, $r^{-2} - 1 \approx 2(1 - r)$ and $\ln(1/r) \approx 1 - r$, so the general formula tends to $\dfrac{2k(1-r)}{2(1-r)} = k$, matching the $r = 1$ answer.
- k is dimensionless (both v^2 and gd are speeds squared), as n must be.
- Stepping the motion numerically for random μ, d, r and v confirms the recurrence to high accuracy.

Common slips

- Applying restitution to v_i^2 with factor r rather than r^2: the law of restitution is about speeds, so squared speeds pick up r^2, and so does the friction term.
- Forgetting that n counts traverses, not collisions with nonzero rebound: the condition is $v_n = 0$, with $v_0 = v$.

Beyond STEP

The recurrence $w_{i+1} = aw_i + b$ is the simplest linear difference equation, and the trick of subtracting the fixed point is exactly the method of "complementary function plus particular solution" you will meet for linear differential equations. Bouncing problems like this (a ball bouncing on a floor is the classic) are also nice examples of energy dissipation and of infinitely many events in finite time when there is no friction. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions and friction with more depth.

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